Sam Gunn

CONTACT

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POSITIONS

Massachusetts Institute of Technology
C.L.E. Moore Instructor of Mathematics (Summer 2026 to Present)

Institute for Responsible Superintelligence (RESI)
Researcher (September 2026 to Present)

Google Research, New York, NY
Student Researcher (January 2025 to July 2025); host: Pritish Kamath

Google Quantum AI, Los Angeles, CA
Student Researcher (June 2024 to September 2024); host: Ramis Movassagh

EDUCATION

The University of California, Berkeley
Ph.D. in Computer Science (Fall 2020 to Present; expected Fall 2026)
Advisor: Umesh Vazirani

The University of Texas at Austin
B.S. in Mathematics (Fall 2016 to Spring 2020)
Advisors: Scott Aaronson and Cameron Gordon

RESEARCH

Manuscripts

1. Sam Gunn. "How to sketch a learning algorithm." 2026. arXiv: 2604.07328. In submission.
2. Sam Gunn and Niels Kornerup. "Review of a quantum algorithm for Betti numbers." 2019. arXiv: 1906.07673.

Conference publications

1. Miranda Christ, Noah Golowich, Sam Gunn, Ankur Moitra, and Daniel Wichs. "Improved pseudorandom codes from permuted puzzles." *Proceedings of the 58th Annual ACM Symposium on Theory of Computing (STOC)*. 2026. DOI: 10.1145/3798129.3800916. ePrint: 2025/2222.
2. Sanjam Garg, Sam Gunn, and Mingyuan Wang. "Black-box crypto is useless for pseudorandom codes." *Proceedings of the 23rd Theory of Cryptography Conference (TCC)*. 2025. DOI: 10.1007/978-3-032-12290-2_7. ePrint: 2025/1021. Outstanding Paper Award.
3. Sam Gunn, Xuandong Zhao, and Dawn Song. "An undetectable watermark for generative image models." *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025. arXiv: 2410.07369.
4. Xuandong Zhao, Sam Gunn, Miranda Christ, Jaiden Fairoze, Andres Fabrega, Nicholas Carlini, Sanjam Garg, Sanghyun Hong, Milad Nasr, Florian Tramer, Somesh Jha, Lei Li, Yu-Xiang Wang, and Dawn Song. "SoK: Watermarking for AI-generated content." *Proceedings of the 46th IEEE Symposium on Security and Privacy (S&P)*. 2025. DOI: 10.1109/SP61157.2025.00178. arXiv: 2411.18479.
5. Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, Yevgeniy Dodis, and Sam Gunn. "Ideal pseudorandom codes." *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*. 2025. DOI: 10.1145/3717823.3718309. ePrint: 2024/1840. Invited to the SICOMP Special Issue for STOC 2025.
6. Sam Gunn and Ramis Movassagh. "Quantum one-time protection of any randomized algorithm." *Proceedings of the 45th Annual International Cryptology Conference (CRYPTO)*. 2025. DOI: 10.1007/978-3-032-01878-6_11. ePrint: 2024/1798.
7. Sam Gunn, Yael Tauman Kalai, Anand Natarajan, and Ági Villányi. "Classical commitments to quantum states." *Proceedings of the 57th Annual ACM Symposium*

on Theory of Computing (STOC). 2025. DOI: 10.1145/3717823.3718264. ePrint: 2024/606. Presented at *Quantum Information Processing (QIP 2025)*.

8. Miranda Christ and Sam Gunn. "Pseudorandom error-correcting codes." *Proceedings of the 44th Annual International Cryptology Conference (CRYPTO)*. 2024. DOI: 10.1007/978-3-031-68391-6_10. ePrint: 2024/235.
9. Miranda Christ, Sam Gunn, and Or Zamir. "Undetectable watermarks for language models." *Proceedings of the 37th Annual Conference on Learning Theory (COLT)*. 2024. ePrint: 2023/763. Presented at *Real World Crypto (RWC 2024)*.
10. Andrea Coladangelo and Sam Gunn. "How to use quantum indistinguishability obfuscation." *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: 10.1145/3618260.3649779. ePrint: 2023/1756.
11. Thiago Bergamaschi, Louis Golowich, and Sam Gunn. "Approaching the quantum Singleton bound with approximate error correction." *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: 10.1145/3618260.3649680. arXiv: 2212.09935. Presented at *Quantum Information Processing (QIP 2024)*.
12. Sam Gunn, Nathan Ju, Fermi Ma, and Mark Zhandry. "Commitments to quantum states." *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC)*. 2023. DOI: 10.1145/3564246.3585198. ePrint: 2022/1358. Presented at *Quantum Information Processing (QIP 2023)*.
13. Sam Gunn, Doseok Jang, Orr Paradise, Lucas Spangher, and Costas J. Spanos. "Adversarial poisoning attacks on reinforcement learning-driven energy pricing." *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*. 2022. DOI: 10.1145/3563357.3564075.

Journal publications

1. Roozbeh Bassirian, Adam Bouland, Bill Fefferman, Sam Gunn, and Avishay Tal. "On certified randomness from Fourier sampling or random circuit sampling." *Quantum* 10, 2002 (2026). DOI: 10.22331/q-2026-02-10-2002. arXiv: 2111.14846.
2. Miranda Christ, Sam Gunn, Tal Malkin, and Mariana Raykova. "Provably robust watermarks for open-source language models." *Transactions on Machine Learning Research* (2026).
3. Sam Gunn, Nathan Ju, Fermi Ma, and Mark Zhandry. "Commitments to quantum states." *SIAM Journal on Computing*, Special Issue for the 55th Annual ACM Symposium on Theory of Computing (STOC). In press. ePrint: 2022/1358.
4. Scott Aaronson and Sam Gunn. "On the classical hardness of spoofing linear cross-entropy benchmarking." *Theory of Computing* 16, Article 11, pp. 1–8 (2020). DOI: 10.4086/toc.2020.v016a011. Presented at the *Twenty-Second Annual Southwest Quantum Information and Technology Workshop (SQuInT 2020)*.

Coverage

- Samuel Greengard. "Can Generative Watermarking Make a Mark?" *Communications of the ACM*, December 29, 2025.
- Anil Ananthaswamy. "Watermarks and Pseudorandom Codes." *Polynomial Times*, Simons Institute, 2025–26, Issue 1.
- Teresa Brooks-Mejia and Christopher Patton. "An early look at cryptographic watermarks for AI-generated content." Cloudflare Blog, March 19, 2025.

Awards and funding

- Lambda Research Grant, 2026
- TCC Outstanding Paper Award, 2025
- Google PhD Fellowship, 2022–2025
- UC Berkeley Lotfi A. Zadeh Prize, 2024–2025
- UC Berkeley EECS Department Merit Award, 2024
- UC Berkeley EECS Departmental Fellowship, Fall 2020

TEACHING

- MIT, 18.02: Calculus. Recitation Instructor (Fall 2026).
- UC Berkeley, CS 172: Computability and complexity. Graduate Student Instructor (Fall 2024).
- UC Berkeley, CS 170: Efficient algorithms and intractable problems. Graduate Student Instructor (Spring 2023).

TALKS

"Tracing data in AI"

- EPFL Confluence Workshop — October 2026

"How to sketch a learning algorithm"

- NYU Center for Data Science Seminar — October 2026
- MIT Statistics and Data Science Seminar — September 2026
- Invited talk at ITC 2026 — August 2026
- Flatiron Institute — June 2026
- The Role of Theory in Trustworthy & Interpretable AI, STOC 2026 — June 2026
- MIT ML+Cryptography Seminar — May 2026
- Institute for Advanced Study — May 2026

"On undetectable backdoors"

- UT Austin Theory and Alignment Group — November 2025

"Pseudorandom error-correcting codes"

- TCC 2025 — December 2025
- Harvard Theory Seminar — September 2025
- NYU Cryptography Seminar — April 2025
- CRYPTO 2024 — August 2024
- UCSB Theory Colloquium — June 2024
- Berkeley Theory Lunch — April 2024
- Google Research NYC — April 2024
- NTT Research — March 2024

"Undetectable watermarks for language models"

- NYU Cryptography Seminar — September 2023
- Berkeley ML Theory Seminar — September 2023
- MIT Cryptography Group — August 2023
- Google Research Mountain View — July 2023

"Quantum one-time protection of any randomized algorithm"

- CRYPTO 2025 — August 2025

"Quantum indistinguishability obfuscation"

- University of Ottawa Quantum Seminar — April 2024
- CIQC Spark Talk — March 2024

"Approaching the quantum Singleton bound with approximate error correction"

- Simons Quantum Colloquium — December 2022

"Commitments to quantum states"

- STOC 2023 — June 2023

"Unclonable cryptography"

- Guest lecture for Quantum Complexity, UC Berkeley — December 2022

"Certified entropy from random circuit sampling"

- Berkeley Theory Lunch — March 2022

SERVICE

Program Committee, ITCS — 2027
MIT Computer Science UROP Mentor — Summer 2026
Berkeley Mathematics DRP Mentor — Spring 2021

REVIEWING

STOC, FOCS, SODA, CRYPTO, EUROCRYPT, TCC, NeurIPS, ICLR, QIP, Annals of Statistics